

ActInf Textbook Group ~ Cohort 2 ~ Meeting 16

(Chapter 7, part 1)

Source: [YouTube](#) Playlist: Uncategorized Duration: 55:55 | Uploaded:
Unknown Transcript method: auto_caption

hello it's February 15 2023 were in cohort 2 of the octave textbook we are in our first discussion on chapter seven so let's just first look at how chapter 7 is anticipated in the text then any of you are welcome to give some thoughts or just raise any thoughts or questions on chapter seven and then that will go wherever it goes we can also look at previously submitted questions as well as walk through the text and get a big picture of what's happening so the first mention of chapter 7 is in chapter one when they're summarizing chapter seven and they describe that in chapter seven they're going to discuss active inference models addressing problems formulated in discrete times as hidden Markov models and partially observable Markov decision processes and they mention the examples they're going to use which is a left or right at A T maze and they also bring in topics information seeking learning novelty seeking they talk about how preference represented by the C parameter is going to be unpacked in chapter seven in chapter four they talk about how message passing will be unpacked further in chapter seven in chapter five they talk about e and policy so as they're mentioning a bunch of the parameters e P_i C message passing they're describing that they're going to come back to it in um if we want to change the way the signal is interpreted we need to rely on learning we will return to this in detail in chapter seven in chapter 7 we will discuss a biologically plausible example of factorization specifically the what and the where streams and in chapter 6 in the recipe they talk about the updating in the form of the updating so several more mentions of chapter seven in chapter six and then we actually get to chapter seven itself all right so anyone just to begin with what were your thoughts or feelings on reading chapter seven or what came to mind as you read or remembered reading chapter 7 active inference in discrete time just on me and go for it yeah well I I didn't read it completely but there's a number of questions that popped up in for instance in figure seven three um so the a matrix is conditioned not only on one state but a couple of states is that correct I know I mean it's uh so the the sequence of preceding States uh takes uh well is there place a role that's different to Hidden

Markov model right and then so this would be one question how this could be and then my another question is whether this pie actually makes this whole automaton more powerful or is this just a convenient way of uh of representing it so um so could this pie actually be merged in into the state as an additional factor or something so multiply the number of states and or does it really increase the computational power of that whole automaton great questions all right so yeah yeah Jacob go for it first I think that the in terms of putting the pie into the states themselves I I don't I don't think I've seen that ever being um done in an actual model and I think this is more of a representation rather than really how the model would be implemented because the pie is just is just the policy and it's I think it's uh it like branches out to the two the two B's uh from like a single node only to represent that each B is also conditioned on the policy uh but not necessarily that it's like a one that that it happens within like one time snapshot yes yeah in in some other work Jacob and I have talked a lot about like why is pi shown to intervene twice and even more generally why is it shown with three time steps yes it helps us understand that it's like past present and future but the action perception Loop also could be shown with just the transition between two steps and Pi intervening once whereas this is a little bit ambiguous whether it's a policy of length two intervening at this time point or whether it's just showing the continual unfolding of policy interleaving between the hidden States in a sort of open-ended way but what what do you mean Michael by increasing the computational power um I mean it it would be possible to have the uh dynamics of the Hidden States through b complex but by removing pi and the action selection apparatus then if we only had the rake down below it would just be a passive inference uh algorithm with a potentially complex endogenous dynamic well I I had the can I give an example yeah oh yeah so I I was thinking of so there is that some kind of recursion possible for instance I was thinking of the following example assume that I want to go and buy a bread and then I go out and then on my way to the bakery I see a haircut and I think oh before buying this bread I could go to the Hakata and then crossing the street I see um um uh a shop with journals and then I think oh when I sit there with a haircut I should maybe buy a journal to read something then I go to that journal I see a friend and the friend tells me oh I have a new car you want to come to see my car and so I passed the half of the day going forward and then eventually I backtrack coming back from the car going by going to the haircutter buying the bread and in the evening I go home so this is a kind of a recursive thing right so so there is an action and then it there's an intervention with another action but then there is uh so it's a kind of a recursive thing and each of those actions maybe follows a policy but that can be there there can be another policy

that inter intervenes but then I backtrack from there so that would actually be um yes a recursive a kind of a policy is is something like this possible so this would really I guess um increase the computational power with respect to just completing one policy or one sequence of actions and then going to the next where you know where one iterates the stuff instead off the possibility to do that recursively so there's something like this I was thinking so is it possible to have some recursive policies there one question I mean with no is to to live stream 42 which was on Robotics and the slam simultaneous localization and mapping there's a higher level policy selection around what location the robot wants to go and then a lower level tactical level that is about the actual motor actuation so in that example that you gave there might be like a higher level model of which location you want to go towards and then there's a lower level footstep model and then this decision this shifting of decision could be modeled as policies on which location one is seeking so as with many features it's not necessarily intrinsic to this figure 7.3 as the kernel but you can compose it in order to capture some of those Dynamics Ali and then Jacob it uh yeah well actually I think uh uh described in those terms I mean uh recursively updating uh the generative model uh is just uh for the it can just be for the purpose of building a comprehensive uh generative model I mean as comprehensively as we can but when we include the policies as the additional parameter in our generative model uh actually this is a kind of um a technique used for bring the elements of action selection uh two or um within the the Bayesian belief updating which is called the planning as inference so in order to do that uh policy selection or I mean constructing a space of all possible actions uh is a necessity a requirement to do to do these uh kinds of planning as inference but yeah it's just a kind of uh mathematical uh trick if you will uh to somehow bring together uh the action selection into the mechanism of uh Bayesian belief updating but but this this possibility oh sorry continue yeah so so this possibility is not in picture seven three right so because these arrows go from PI to the B Matrix but there's not a kind of an error going back so so if I are you saying the trick would be that one learns a pie inside a pie somehow yeah you see this is just one policy a picture a snapshot of a one policy uh but uh in the more comprehensive models we have uh the state of possible and I mean uh we have policies and not just one single policy to select from so yeah that's just that picture just shows how uh the uh I mean choosing a single policy can affect uh the uh generative model and how it can do the inference via planning or action selection but in reality or I mean uh when dealing with more comprehensive uh models we have uh not we don't necessarily have just one single policy to do uh to uh select from I mean uh it would be tautological to have just one single policy so we

need a space of uh possibility yes if there was just one policy and there is no agency around selecting then it reverts to a passive inference challenge whereas the structure of the B Matrix which you can explore from a Hands-On perspective like by looking at the dimensionality of it in piodp the inference problem is set up such that agents are deciding between π_1 and π_2 which are evaluated according to their expected free energy and π_1 and π_2 do different things to be so we can think of like a B1 and B2 not time point one and two but variant or slice one and two that change how hidden States change their time so Jacob I guess on the question of um like recursion and uh with your example of like going to to a barber and then seeing oh um I want to buy a magazine to read and like going the policy changing throughout I guess I'm not uh I'm not really sure I see the recursion there in in like an active inference model because I think it's more I'd say it's more like a very high dimensional model where you are perhaps performing uh where you're always like narrowing down your policy search space and with new observations you're like adjusting that search base based on your preferences in that particular uh observational modality like you get reminded that there are magazines and that you can buy magazines to read uh while you're at another place so it kind of extends that computation but I don't think it's necessarily um recursive and also I think the the figure um I think you mentioned something about like uh prediction errors or uh so something like that I don't think that's necessarily captured in the figure because that only shows well I I guess it's not explicit but it we can assume that the B There is static and you can imagine a lot of other things happening if you would be learning the B major fix or doing uh model selection as well um and adjusting your policy based on what observational modality has the highest um precision and can I answer uh yeah to this so so I think well maybe you can represent this in a in a way of an iteration or the way you say but it it would be recursive in my understanding of the these terms if if going for instance to the baker is is a policy means it's a series of actions right and so if this is a policy and then in the middle of this there is somehow another policy that pops up oh I go first to the hair cutter or to the barber whatever and then this uh one policy still exists but inside there's another kind of sequence of actions another plan and so on popping up and this if this could happen recursive I mean in my understanding of this term of um of these terms uh it it would be kind of recursive no yeah yeah it can be through nested modeling which is one canonical way that recursion is implemented so it's it's unfolding sequentially Through Time but also like this higher level um discrete time so that's again we're chapter seven we're in discrete time and so this might be a slower click so maybe the footsteps are taking place on a second to Second policy tick timeline whereas a policy in which location

one wants to go is being evaluated at a slower level and if this nested level is playing out at a similar frequency to the lower level than as a modeler you might be better off just having it as a single level model and as an organism you'd get into these kind of like paralysis cases where if you're actually switching policies about where you want to go then this would be incoherent down a lower level so just return to the initial question is it possible to merge policy into the hidden States as a causal latent factor well in a way policy intervenes in how hidden States through change their time so policy does influence hidden States without an action selection apparatus it's not active inference it's not even a control theory problem it's just a question of identifying latent causes from the unfolding of a hidden States and observables that you can't control and that's a classical SPM type challenge like s might be the latent unobserved neural activity and O might be multi-sensor fusion with fmri EEG and Meg so this can still be very complex in its unfolding through time with the observations in the hidden States what would make it an active inference problem well if you weren't just observing sensor modalities but if the experimenter could take a policy that was intervening how neural Dynamics change through time which is what experimental design is which is why it's so compatible with thinking about the metabasian approach with the behavioral researcher so apply influences hidden States but if we merely folded into B we would lose the ability to do actual action selection but if it were a passive inference problem we still could use variational inference we still could use a free energy we still could even align outcomes with our preferences we could still calculate and bound surprise the active embedding of active inference is to include action as a parameter like Ali mentioned within the Bayesian brain context to have a parameter that you do Bayesian inference on and then extend that not just from perception and causal inference which is what's happening on the bottom like perception is the a causal inference can be understood as B but action selection does require an additional apparatus passive inference more like signal processing action more like control theory unified imperative so the G would then be predicting the future of expectations developing of expectations and so on what would it be what is g the G the π oh sorry oh well π is the combinatorics of affordances over a given time Horizon so merely generating π as a list is not intrinsically difficult it's just the search space of possible action sequences over a given time Horizon if there's two options and you're doing a Time Horizon of three then it's two cubed um G is the expected free energy which updates the policy prior so take that list and it's your prior probability of those policies which could be either just a uniform prior or could reflect habit like a priori How likely are policies and then G sharpens or updates that policy

vector into a policy posterior which then the agent samples their actions from and and how um how is this meant so I assume that I'm so you're saying the first is the past and then um so is this is this re re-estimated at every state so so the Spy would somehow have um expectations of a future affordances as you say right but would say which which transitions are more likely in the uh to the right of this graph and would that be re-estimated in at each time time step or okay how would that yeah there's there's multiple possible levels only go first then I'll give a thought well basically Π actually is just an additional parameter uh multiplied to the basic uh Bayes theorem formula we had before so instead of just multiplying likelihood and uh the priors simply we also multiply the probability of selecting a policy and that's just it so Π is just an additional parameter in that equation from chapter back in chapter two yeah equation 2.6 is where G is defined so G is a a function of policies so it takes in as arguments the vector of policies and then conditioning as we see in every single of these phrasings every single one is conditioned upon a given policy and then different policies are compared by their expected free energy and then depending on the Precision on the action posterior which isn't shown in 7.3 you can have a spectrum of behavior if you have extremely high precision like a um it could be thought of as low temperature in that case one policy that's let's just say one is um it's one percent better than the other then under the low temperature asymptote the high Precision asymptote the policy that's even tiny tiny bit better will always be selected under the neutral temperature Zone policies are selected based upon their posterior likelihood so if one of them is 51 percent and the other is 49 when you soft Max it then it's selected 51 and 49 of the time respectively and then in the high temperature or low Precision asymptote also known as shaky hand the differences amongst policies in their expected free energy basically get erased and at that point policies are selected essentially without respect to their posterior likelihood this is one of the key pieces though because again we're not calculating different policies and asking well how rewarding are they how much expected reward do they have and then ranking or selecting based on that rather there's a unified imperative expected free energy which under different decompositions can be shown to have these different interpretations that is then used to generate a policy posterior and then as just described that policy posterior in the context of a of a temperature or Precision variable can exhibit behaviors ranging from always selecting the slightly preferred policy in the low temperature limit neutral temperature selecting according to the posterior high temperature erasing differences in the posterior Ali yeah uh sorry I just wanted to point out that uh this G Matrix here is uh still prior to learning so uh if we want to add learning to uh to into the policies uh things would get a little bit more uh not to say

complicated but more extended equations would be needed so uh we haven't included learning in this specific formulation yet it's prior to learning yes thanks yeah I I um I also want to highlight that like in in the category of phenomena that we can model but are not just simply in this image you mentioned like learning new policies so it's like in the fixed fixed fixed case you're still able to do action selection but you're not learning or changing which policies you can enact you're not changing what affordances you believe you have nor are you updating B which is to say you're not updating your beliefs about how policies influence hidden State changes the next level of introduction of novelty is learning on B where you can learn about the different consequences of policy and then another level which is starting to approach into like the structural learning because you're actually changing not just parameters within the model but dimensionality of parameters within the model is like well what if I take a policy that opens up another set of possible policies so this is like see the cat run of sentences and to introduce recursive or referential Clauses or learning it's a more complex sentence so to speak but this is the archetype that sets the stage for what cognitive model structures could exhibit that kind of a phenomena Michael yeah so um a policy would be a number of actions right would be a sequence of States yeah is is that am I understanding correct and if this is the case then a policy would select a number of states and my question I guess is um so then I come at the end or at what point are these policies then uh calculated so I come to the end of my uh of a of an action and then only is the uh it's the it's a new um policy uh a computer or selected or is it constantly the case so if um if I decide to to go to buy a bread for instance and then I go out and I cross the street and do this and that so there's a couple of actions involved right and so my decision to buy a bread was done before but on the way um is that policy recalculated and maybe on on the way I think oh I um uh something else is is more reasonable to do okay so again that's it's a slightly nested scenario with with multiple levels of policy that you're um looking at and then just to be clear so policy it's a sequence of actions so let's just say that the affordances there's two affordances up or down and then time Horizon is three well let's just say it's two there are four policies you you d d u d d so generating the state space of policies is trivial it's just all of the strings of affordances now you ask where does policy inference happen where and how does it happen and there's a few different answers so first off here's from our active inference and modeling conflict paper here we chose to show only a two-step model so instead of the the three because we were anticipating that um ambiguity where P_i is intervening twice when actually control is only exerted at any given moment but it's it's anticipated um and then we map this onto the UDA observe Orient decide act this isn't a

formal mapping per se but it helps understand sequentially for example in a reactive message passing paradigm what different informally categorized states of the perception cognition Action Cycle how that maps onto this basing graph so since States come in cognition on occurs or one could also take a broader perspective then but this updates one's beliefs about the world then there's an inference on action so here are the antecedents of policy selection which are e what can be done the affordances see what is preferred the preferences preferred slash expected and then with c and e as it shows here G is a function of policies conditioned upon c and e so with those antecedents taken care of we can update our beliefs on π_i to move from a policy prior to a policy posterior those policy posteriors can then be sampled from as discussed earlier at high low or neutral temperature and then the selected policy intervenes and ends up choosing a slice in b and then the slice of B that's chosen influences how the hidden States change for example if there's two Paul if there's two policies on a one-time step basis so same as the affordances in this case turn on the heater or do not turn on the heater so maybe if we don't turn on the heater then the B Matrix just passes the hidden State forward the temperature doesn't change if we do turn on the heater the temperature increases immediately again simple example we looked at the thermometer oh updated our belief given our priors about the temperature of the room then with respect to what we can do e and what we expect slash prefers C we updated our policy prior into a policy posterior we then selected the action that's most likely not the most rewarding action the most likely action and that influenced how the hidden States changed their time and then the UDA cycle begins again so action intervenes in the discrete time formalization which is where we're in chapter seven it intervenes in every time click okay thank you so much and and also um the pragmatic turn is um it's a book co-edited by Friston but it discusses like what has happened in Neuroscience with people embedding action and thinking about action in neuroscience and that's relatively recent which is really exciting okay but these are these are great questions and then let's just also go to the um second question that you raised there and then we'll scan through all of so is the a matrix in figure 7.3 conditioned on one state or a series of States so Jacob feel free to add any more details on this one but as it's shown here from a base graph perspective it is O of T minus 1 is only conditioned on S of T minus 1. but when we think about across the vector of observations they can be understood as being conditioned upon the time series of hidden States so each observation is only conditioned upon its Associated hidden State that's the partially observable and hidden States pass through the Markov needle of the presence that's the markovian property past only influences the future through the

present which isn't to say that's how reality is that just to say it's a simplifying structural assumption that one could vary if they chose but it is a little ambiguous I I I can see why somebody might think that observation at a given t let's say t plus one is can is a function that's conditioned upon all the previous types but not directly observations are like cul-de-sacs and again a is The Tail of Two densities the recognition density which is from observations updating beliefs about Bayesian recognition model and the generative density which is from a hidden state generating pseudo-observations and the way that those two densities can be alternated is used and has been used for decades in expectation maximization algorithms okay cool great questions here does anyone want to go to any of the chapter 7 questions or should we scan through in the last 15 minutes here just like what happens in chapter seven foreign Let's scan raise your hands when you want and I'll immediately go to you if you want to um just add a note on like as we scan through so we will scan through the whole thing in the next 10 minutes okay starting with a quote what I cannot create I do not understand famous quote take it or leave it but now we're into the building the recipe has been disclosed in chapter six the prerequisite knowledge has been distributed in chapters one through five ideally they start with perceptual processing this is very much like the step-by-step paper step by step starts even simpler with starting with just a static inference but here they go immediately to Dynamic perceptual inference this is just the bottom of the rake this is a passive inference but it is changing through time they're going to be using a music example speaking to their target audience which is Ali um a musical example so we're listening to some music not necessarily playing um the music but the outcome is the note that's actually intended I'm sorry the outcome is what is actually played and the hidden state is the intended note and just from looking at this a matrix we can see that like on the diagonals if it was the identity Matrix that would mean that there was a perfect mapping between hidden State and observations but this is not an identity Matrix it has off diagonal values which is to say it's like a blurry a matrix and in the limit of a blurry a an ambiguous a you could imagine if it was all 2.5S across then there would be no information on hidden State given the outcome The Other Extreme would be perfect information with the identity Matrix and you can see that it's also rescaled so the numbers can be shown kind of Simply as integers and then it's rescaled so that it's a proper probability so seventy percent of the time the musician hits their intended note Row one column one seven B is the probability of transitions so we can go if we go from Node 1 to Note 2 with 97 so notice the diagonal if that was the identity Matrix on B if that was all B was you would basically dwell wherever you began if b is all off diagonal you never stay in the

same spot twice and so here is a sequence of notes because one almost always moves to two two almost always moves to three three almost always moves to four and four almost always moves to one then D is the prior and so it's just being taken as a noiseless estimate that the first note is the beginning note Michael so is this a very boring tune that plays Four notes in the in the round eternally exactly like this it's like Geb in Eternal Golden Braid except it's like g e b x g e b x okay very boring yeah okay yes it is a simple one so 7.1 to 7.3 equations completely specify the hmm note that it's a hidden Markov model because it has this partially observable character but it's not a decision process because there's no action selection but all we needed to do was to specify a b and d so to totally specify the hmm was ABD we didn't have to say which notes we preferred we didn't have to say anything about what we could do with C or E then there are some simulations shown um there's a few things about this simulation first as the earlier question has raised why are there continuous interpolations when we're in the discrete time world a general comment is these graphs with multiple block lines are a little hard to interpret because of course we can't tell which line is which another issue with 7.2 figures specifically is the bottom left quadrant is empty so just a few points but what's basically happening is in this simulation run which would only happen this way um several percent of the time but they're showing it um basically um in the upper right the retrospective beliefs are shown so the retrospective beliefs are shown with a correct song even though the actual observations were were note one two two four so this observation was recognized and it did increase the posterior on Note 3 being played however because the belief was so strong that belief that Note 3 would be played in this time it did not move the needle far enough to actually hear that note so that's like when somebody hears something that isn't said because it's likely that it was set okay so that's a fun model but it's not action inference yet so that's where we get to 7.3 where we've been spending a lot of today's discussion 7.3 discusses the addition of an extra variable on which transition probabilities are conditioned so it's like really when we think B we should really think B vertical line π because the inference on action means that if that we have actual options for action which is to say there are multiple B's that can influence how s unfolds so we're really talking about B's condition on on policy unless we're talking about like the whole B caterpillar the whole tensor of B here's where um expected free energy and one of the partitionings is shown which is discussed in chapter two equation 2.6 and in chapter four so next week when we're in chapter seven for the second time let's can let's return talk a little bit more about the technical parts of like what exactly equation 7.4 means it's a decomposition that enables epistemic drive towards information gathering that's this first

part of the right hand side and the pragmatic Drive which is about realizing prior beliefs
 reducing surprise about prior beliefs not maximizing a reward function reducing diversions
 between preferences and outcomes they're going to unpack the full 7.3 figure in the context of
 a t maze which is a rat that's able to go left or right to find food or not or it's able to get a cue
 that's the Q location r there's four locations that it can be in and there's modular degrees of
 freedom in how different things are represented so this isn't even the only way to represent this
 toy example but this Matrix does a really good job of showing like what are the states what are
 the dimensionality of the difference variables which we can come to next week so here's
 context two here's context one here we have block on the right side white on the left here we
 have block on the left white on the right and there are some subtle differences which you can
 note like specifically with lines two and three here's a one here here's a one on the third line
 because the Q is suggesting that it's on here the queue is showing it's on the left and in model
 stream 7.2 that example is like very Hands-On and in some ways even better than this one and
 it includes things like learning but it's a very similar task here is B2 so they're showing the
 dimensionality of what those B slices look like for different policies this is representing the B
 tensor here are preferences this is what a preference Vector looks like and in step by step the
 paper and model stream one series and in model stream 7 Series there's some discussion on
 what does c mean is it being reflected as log probabilities why would one do six and negative
 6 versus 6 000 a negative 6000 or 0.6 and negative point six what if those values actually
 mean and the prior is provided so now that we've provided a controllable B which is the
 manifestation of policy options and provided a c we can do active inference because we have
 now what we need those two pieces really that's what brought us from the hmm to the pomdp
 was what we can do and what we prefer um section 7.4 goes into information seeking it
 unpacks the epistemic value into a few more sub decompositions and it talks about how that
 plays out there's some Nuance here again there's also some semi-confusing black and white
 graphs but other ones are clear we can talk about how epistemic value is being realized in this
 example box one is a a bit of a uh side note on uncertainty and precision and here's showing
 really the temperature or the Precision highest lowest entropy most uniform distribution lowest
 temperature highest entropy most specific distribution here you're sharpening here you're
 blurring okay 7.9 as mentioned um two pages earlier is an icate there are four locations and
 the eyes are cicading again one could ask about where are these continuous interpolations
 coming from since we're in chapter seven in a fully discrete situation but it's all good so
 isocating different sequences or traces of isocades across four locations driven by epistemic

value in section 7.5 learning and Novelty is introduced so as we were mentioning earlier learning with respect to the a or the B Matrix so far we've been talking about them as fixed here is where learning happens on a and b and just like you would do with any Bayesian inference whether on action or on the consequences of action or on perception learning is modeled as a hyper prior or a variable that governs a prior on another parameter so here is like a parameter u_m that is adding a layer to the rake we still see the core figure 4.3 figure 7.3 Motif but there are these new little edges that are the learning edges so it's the same pumpdp as figure 7.3 but the priors for each hidden States now depend on the variables which themselves come with prior beliefs so the prior on the variables A and B is in the Bayesian Paradigm how we learn or update our values on a and b brief discussion on College of priors some more equations that we can come to next week about updating and learning on a another look in equation 7.11 on G which is very similar to what we saw in equation 2.6 different partitionings on expected free energy a policy then there's a final example that's now in a maze and so this is some maze seeking entity we can come to this next week u_m in hierarchical inference hierarchical or deep inference something we kind of mentioned today but we'll look more at next week is u_m well first there's a box on structure learning which as mentioned earlier is when we think about look regular non-structured learning as like fine-tuning quantitative parameters within a given dimensionally defined generative model structure learning and what you can do with structure learning like specifically Bayesian model reduction is when we're actually modifying the dimensionality of the generative model that's in box 7.3 here is a nested generative model which is almost exactly the form that we looked at from live stream 42 where we see like the rake classical figure 4.3 figure 7.3 Paradigm and then there's like a meta break where here's another E downward facing and we have policy at a higher level and different generative models have taken different interpretations on what the lower and the higher level are like in live stream 42 the lower level was tactical movement of a robot and the higher level was strategic location preference whereas in live stream 28 sandbed Smith at all the higher levels were associated with attention in the lower levels with perception another example with decision making will come to and then the chapter summarizes so thank you for this session we'll stop recording and take a break before the next one